

AMIRALI AALI

M.Sc. Data Science Student

| RWTH Aachen University

SAP · Porsche · Deutschlandstipendium Scholar

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PROFILE

Data Science M.Sc. student at RWTH Aachen with 3+ years of experience in computer vision and data privacy engineering (Porsche, SAP, Aptiv). Specialized in real-time ML systems and deep learning applications in automotive contexts.

TECHNICAL EXPERIENCE

Intern — Development of Vision-Based Driving Assistance Systems

Porsche

📅 Apr 2026 – Aug 2026 • 4 mos

📍 Weissach, Germany

- Currently working on data-centric development of ML models.

Working Student — Data De-Identification

SAP

📅 Apr 2024 – Apr 2026 • 2 yrs

📍 Düsseldorf-Ratingen, Germany

- Part of the anonymization team delivering enterprise-scale data de-identification services.
- Designed and maintained Databricks pipelines integrated with Azure Data Lake Storage (ADLS), implementing Python-based text anonymization workflows compliant with data protection requirements.

Bachelor's Thesis Student

Aptiv

📅 Feb 2024 – Mar 2024 • 2 mos

📍 Wuppertal, Germany

- Thesis: *Street Lamp Area Detection* — a joint project with the University of Duisburg-Essen combining classical vision and CNN inference.

Working Student — Computer Vision / AI / ML

Aptiv

📅 Nov 2022 – Jan 2024 • 1 yr 2 mos

📍 Wuppertal, Germany

- Collaborated with a specialist team on a deep-learning high-beam assistant for Gen. 7 automotive systems.
- Built a real-time smart controller that automatically adapts high-beam intensity based on scene detection.

SKILLS

Data Science & ML

Python PyTorch OpenCV

NumPy Pandas Matplotlib

Scikit-Learn Computer Vision

Reinforcement Learning

ADLS Cloud Development

Databricks

Systems & Low-Level

C/C++ Rust PyBinds Bazel

DevOps & Tooling

Git/GitHub Gerrit Jira

Confluence L^AT_EX

LANGUAGES

English C1

German B2

Persian Native

HONOURS & AWARDS

German National Scholarship

Deutschlandstipendium — RWTH Aachen

Sponsor: Dr. Ing. h.c. F. Porsche AG

2024–2025

3rd Place — National Aerospace Competition

Glider League, Amirkabir University Apr 2016

Tutor — Procedural Programming (C)

Uni Duisburg-Essen

May 2022 – Aug 2022 • 4 mos

Duisburg, Germany

CERTIFICATES

EDUCATION

M.Sc. Data Science

RWTH Aachen University

Apr 2024 – Present

Aachen, Germany

B.Sc. Software Engineering

University of Duisburg-Essen

Oct 2020 – Mar 2024

Duisburg, Germany

Thesis: *Street Lamp Area Detection using Classical Vision and CNNs*

GPA: 1.7 (very good, German scale)

Preparatory College — Freshman Program

FH-Aachen

Oct 2019 – Aug 2020

Aachen, Germany

Reinforcement Learning: Beginner to Master

Udemy

Python for Data Science & ML

Udemy

C++ Programming: Beginner to Beyond

Udemy

RECOMMENDATIONS

"Amir consistently demonstrated a deep understanding of AI and ML, strong problem-solving skills in Python and C++, and a passion for continuous learning. An invaluable asset to any team."

Markus Bühren

Manager, Computer Vision — Aptiv

May 2024 · [View on LinkedIn](#)

"Amir built scalable data-anonymization pipelines and Python workflows, applied research to real-world solutions, and contributed to code quality with CI practices. A quick learner, reliable team player, and strong communicator who consistently delivered high-quality results."

Henning Schmitz

Head of EDT Data Protection — SAP

March 2026 · [View on LinkedIn](#)

"Amirali brings solid technical knowledge across ML, AI, and data privacy, and is equally comfortable tackling diverse problem types. Collaborative, communicative, and a pleasure to work with."

Louay Abdelgawad

Data Scientist — SAP

March 2026 · [View on LinkedIn](#)

PROJECTS

Rapid Time Series Datasets Library with Rust

[Rust](#) · [Python](#) · [NumPy](#) · [Gitlab](#)

Collaborated with a team to develop a Python package backed by Rust, aimed at efficient preprocessing of time series data. The package integrates seamlessly with PyTorch via the LightningModule interface and is designed with performance and memory efficiency in mind.

[View on GitHub](#)

Street Lamp Area Detection

[Python](#) · [OpenCV](#) · [PyTorch](#) · [NumPy](#) · [Pandas](#)

Nighttime street lamp zone detection combining classical image processing with CNN-based inference; bachelor's thesis project.

[As per company policy, code is not publicly available](#)

Face Anonymizer in Rust

[Rust](#) · [Python](#) · [PyBinds](#) · [Computer Vision](#)

High-performance face anonymization pipeline with Python bindings, enabling real-time processing.

[View on GitHub](#)

Rocket Interceptor

[Python](#) · [PyTorch](#) · [Reinforcement Learning](#) · [PyGame](#)

Deep RL agent trained to intercept projectiles in a custom PyGame simulation.

[View on GitHub](#)

Autonomous Car Navigation (PPO)

[Python](#) · [PyTorch](#) · [Reinforcement Learning](#)

Obstacle avoidance using Proximal Policy Optimization.

[View on GitHub](#)